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<u>Título/Title</u> Program FLEXA Version 3 User's Manual				
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<u>Resumen/Summary</u> This document is the user's manual of program FLEXA, version 3. The FLEXA program checks the strength of beams subjected to axial and bending loads. The program takes into account the elastic and plastic behaviour of the beam material. The program calculates: - Stress values distribution on the beam cross-section due to the applied loads and also at the failure loads. - Reserve Factor corresponding to the allowable maximum strain.				
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1 INTRODUCTION

1.1 TECHNICAL SHEET OF THE PROGRAM

Technical Report: ME-T-NT-110142

PROGRAM

FLEXA

DATE

Feb/2016

OBJECTIVE

This program checks the strength of beams subjected to axial and bending loads. The program takes into account the elastic and plastic behaviour of the beam material.

LANGUAGE

Fortran 77

Visual Basic Net

DEVELOPMENT TOOLS

Oracle Solaris Studio

MS Visual Studio 2013

OS/PLATFORM

Unix (Solaris) and Windows (x86)

Windows (x86)

NUMBER OF LINES (APPROX.)

8600 (Fortran, analysis core)

+3600 (Visual Basic .Net)

Total 12200

EXECUTION MODE

Console (input data in text file)

Interactive GUI

Notes:

- Blank lines and commented lines excluded.
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files.

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VERSION

1.0 / 2.0

3.2

3.3

3.4

ADDITIONAL DOCUMENTS

See list of applicable references in sub-chapter 1.4.

1.2 SOFTWARE PURPOSE AND DESCRIPTION

The FLEXA program analyses the internal forces in beams subjected to bending loads or to a combination of bending and axial loads. This manual describes the use of the version 3 of the program, including examples to further clarify its use.

The analysis performed by FLEXA includes:

- The geometric properties of the beam cross-section (area, inertia and the position of its centre of gravity).
- The position of the neutral axis and the distribution of strains and stresses throughout the section that correspond to the applied loads.
- The position of the neutral axis and the distribution of strains and stresses throughout the section that correspond to the failure loads.
- The reserve factor that corresponds to the maximum allowable strains.

It should be remembered that, in general, when the material enters in plastic regime zone, the position of the neutral axis is not the same as its position while the material's behaviour is elastic. This is a consequence of the stress-strain ratios that exist in the plastic regime.

All the loads are considered applied simultaneously, keeping constant the ratio between them. In consequence, the failure loads are proportional to the applied loads, and the reserve factor is the ratio between them.

The cross-section of the beam can have any shape (open or closed, thin-walled or otherwise); as long as it verifies that its contour is composed only of straight lines and/or circular arcs.

The program assumes that sections that initially were plane remain also plane after load application.

The program assumes that the material is homogeneous and the ratio between the stresses and the strains in the material meets the Ramberg-Osgood law, as indicated in the Theoretical Manual. The behaviour of the material when it is subjected to tensile stress may be different to its behaviour under compressive loads. This is common in materials used for aeronautical construction.

The allowable strains corresponding to tensile stresses can be different from those for compression. Either may be greater than the material's limit of proportionality. Moreover, the program's real usefulness is when it is used to study cross-sections in which plastic strains are produced, since while the strains remain within the elastic regime, classical methods for treating elastic bending of beams would be more appropriate, although this program may also be used for such calculations.

Finally, it should be pointed out that the program assumes that the sections are stable, i.e., no local buckling is produced at any part of the section. The user should confirm that afterwards.

1.3 ABBREVIATIONS AND SYMBOLS

<u>Symbol</u>	<u>Units</u>	<u>Definition</u>
DLL		Dynamic-Link Library
GUI		Graphical User Interface
I/O		Input and Output
MSVS		Microsoft® Visual Studio® 2013 (plus Intel® Visual Fortran Composer version 14 and IMSL Fortran Numerical Library version 6.0.0)
OS		Operating System

1.4 REFERENCES

	<u>Document Reference</u>	<u>Issue</u>	<u>Title</u>
Ref. 1	NT-T-ADCE-012/92	A	Theoretical Manual of the FLEXA Program.
Ref. 2	NT-T-ADCE-013/92	A	Programmer Manual of the FLEXA Program.

1.5 DIFFERENCES AMONG VERSIONS

Version 1.0:

Initial version developed in Fortran 77 for Digital VAX computer.

Version 2.0:

The program is adapted to the new Sun Solaris computer.

Version 2.1:

A cover of the program is now written in the output file.

Version 2.2:

The protection method against unauthorized use is changed.

Version 2.3:

The material properties must be directly entered by the user instead of being selected from the material database.

To finalize the data entry while defining the cross section or entering the load cases, the user have to enter a line with an asterisk (*) as its first character, instead of pressing <CTRL/Z>.

The user can choose the language for dialogs on the screen and for the output results file, between Spanish or English.

The program source is restructured to allow the creation of a DLL for Windows OS.

A Windows application has been developed as GUI for the program FLEXA. This application uses the same input file and provides the same output results. The section properties calculation as well as the bending analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

Version 3.0:

The user can choose the language for dialogs on the screen and for the output results file, between Spanish or English. In previous version it was only possible for the GUI.

The program is re-arranged. The format of the input file is different from previous versions of the program, with data structured in groups, in a similar way as other applications.

The output results are now provided in a single file, different from the previous version. The generation of a file with Latex code with a sketch of the section has been avoided.

The GUI is improved. It presents the results on the screen and it is possible to write them in a text file, identical to the output file from the official program in Fortran.

The GUI allows the user to obtain a sketch of the beam cross section in a graphic file.

Version 3.3:

The program allows the using of materials with F_{cy} greater than F_{tu} . For these cases ($F_{cy} > F_{tu}$) the material in compression is assumed perfectly-plastic after yield, and the strength cut-off is set at the point with the same permanent deformation after failure as in tension.

Version 3.4:

Uncontrolled bugs which cause the interruption of program when defining cross-sections by means of coincident points and/or too short corner radius values for the defined section have been fixed for both GUI, Windows console and UNIX console program versions. Additional error messages #18 and #19 accordingly added to the program console versions.

The resulting reserve factors are truncated and shown at both GUI, Windows console and UNIX console program versions according to Airbus DS usual policies.

Other minor improvements are incorporated such as memory of active folder, regional and decimal local configurations, writing of header with basic information of the program at results output reports and program version and active file name at the title of the GUI window forms.

2 PROGRAM EXECUTION

2.1 WINDOWS/PC VERSIONS

An application has been developed in Visual Basic for Windows/PC environment. The original routines of the program for input data, calculation and output results have been included in a DLL, which is used by this application. Therefore, the same results will be obtained, with only differences due to numerical rounding.

The Visual Basic executable program `Flexa.exe` (as well as an electronic copy of this manual and the program examples) is located in the folder:

`\\Cae\InfoCalculo\Soft\Flexa`

To execute the program, the user has to click twice on the Visual Basic executable program.

There is another executable file with the same name, which is the Fortran program in console mode, in the subdirectory `.\Console` within that folder (and another copy of the console-mode executable in `\\Cae\InfoCalculo\Soft\Console`, this last path than can be added to the Windows system PATH to facilitate its use).

2.2 UNIX/MULTIVAC VERSION

Prior to the program execution, it is necessary to create the input data file, as described in section 3.2.

To run interactively this version of the program (console application), you must have already done the assignment of the symbols list in Airbus DS Structural Analysis Office using the command:

```
. asignaciones_calculo
```

This instruction can be executed in the commands terminal prior to execute the program, but it is strongly recommended to include this statement in the files `".profile"`, and `".kshrc"` (if working in Korn Shell) or `".bashrc"` (if you work in Bash Shell). To do this, the following lines should be included in the file `".profile"`:

```
. asignaciones_calculo
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
```

In files `".kshrc"` and `".bashrc"` the following line shall be added:

```
. asignaciones_calculo
```

The instruction to run the program in interactive mode in the MULTIVAC server is:

```
$flexa
```


The program asks for the language to use and the name of the input and output files, as shown in the following example:

```
FLEXA 3.4                      (c) Airbus DS - 2016                      16/Feb/2016
-----
Idioma / Language:
=====
  1 ... ESPAÑOL
  2 ... ENGLISH

Idioma / Language: 2
Name of the input file:  filein.dat
Name of the output file:  fileout.txt
Reading input data file
Calculating section properties
Performing flexion analysis
Writing the output file
FLEXA 3.4. (c) Airbus Defence and Space 2016
```

The CPU time for calculation depends on the number of load cases and it is, in general, negligible.

This console application can be executed in one single command, by passing as arguments the input data file (i.e. filein.dat), the desired name of the output results file (i.e. fileout.txt), and an integer number to select the report language (1 is Spanish, 2 is English):

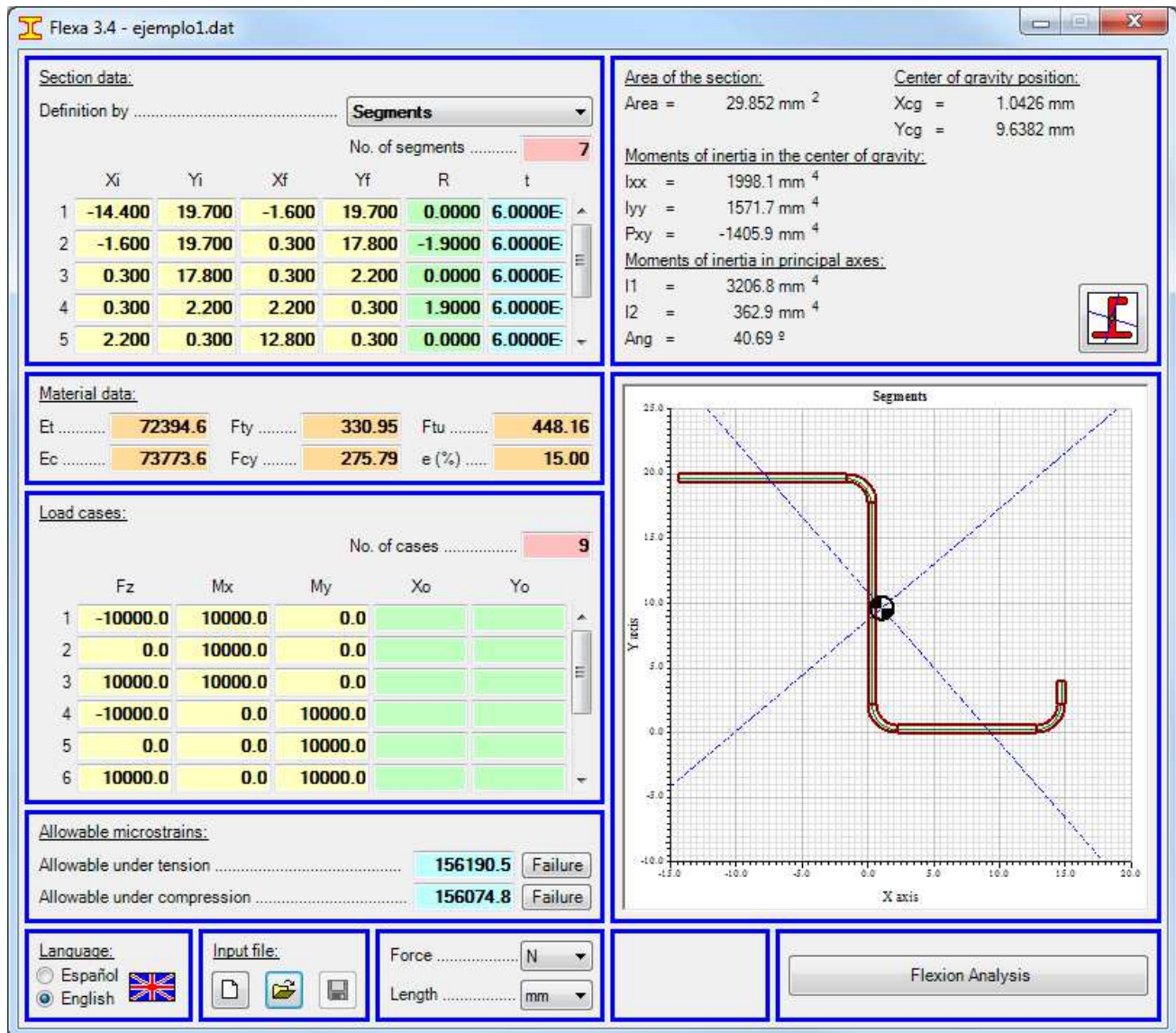
```
$flexa filein.dat fileout.txt 2
```

If the third parameter (language) is omitted, the English language will be used.

3 INPUT DATA

3.1 WINDOWS PC/GUI

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the Example 1 (see chapter 6), has the following appearance:



Flexa 3.4 - ejemplo1.dat

Section data:
 Definition by Segments
 No. of segments 7

	Xi	Yi	Xf	Yf	R	t
1	-14.400	19.700	-1.600	19.700	0.0000	6.0000E
2	-1.600	19.700	0.300	17.800	-1.9000	6.0000E
3	0.300	17.800	0.300	2.200	0.0000	6.0000E
4	0.300	2.200	2.200	0.300	1.9000	6.0000E
5	2.200	0.300	12.800	0.300	0.0000	6.0000E

Area of the section:
 Area = 29.852 mm²

Center of gravity position:
 Xcg = 1.0426 mm
 Ycg = 9.6382 mm

Moments of inertia in the center of gravity:
 Ixx = 1998.1 mm⁴
 Iyy = 1571.7 mm⁴
 Pxy = -1405.9 mm⁴

Moments of inertia in principal axes:
 I1 = 3206.8 mm⁴
 I2 = 362.9 mm⁴
 Ang = 40.69 °

Material data:
 Et 72394.6 Fty 330.95 Ftu 448.16
 Ec 73773.6 Fcy 275.79 e (%) 15.00

Load cases:
 No. of cases 9

	Fz	Mx	My	Xo	Yo
1	-10000.0	10000.0	0.0		
2	0.0	10000.0	0.0		
3	10000.0	10000.0	0.0		
4	-10000.0	0.0	10000.0		
5	0.0	0.0	10000.0		
6	10000.0	0.0	10000.0		

Allowable microstrains:
 Allowable under tension 156190.5 Failure
 Allowable under compression 156074.8 Failure

Language:
☐ Español 
☒ English 

Input file:

Force N
 Length mm

Flexion Analysis

Segments
 Graphical representation of the section geometry on an X-Y coordinate system.

The program automatically recalculates the section properties at the time that section definition data are changed.



This button, located at the upper right frame, obtains a draw of the section in a graphic file.

Values shown in the input form have the same meanings as in the input text files. Detailed explanations can be found in following sub-chapter.

3.2 INPUT FILE

The input data file must contain all the necessary data that defines the section, the material properties, the load cases, the allowable strains and the units for lengths and forces.

It basically contains data groups, numbered from 1 to 6, appearing in any order. Some of them are optional and groups 1 and 2 cannot be together. The content of these groups is described as follows:

GROUP 1: Section defined by segments:

This group contains the geometrical definition of the cross section of the beam when it is defined by segments.

If the group 1 is present, group 2 have to be omitted, but one of them must be present.

The structure of this group is the following:

Line 1 -----> 1 "Optional commentary"

[Ns] lines ----->	Ns	Xi(1)	Yi(1)	Xf(i)	Yf(i)	R(1)	t(i)
		Xi(2)	Yi(2)	Xf(2)	Yf(2)	R(2)	t(2)
							
							
		Xi(Ns)	Yi(Ns)	Xf(Ns)	Yf(Ns)	R(Ns)	t(Ns)

Where: Ns = Number of segments to define the section.

Xi(i),Yi(i) = X and Y coordinates of the initial end of the segment.

Xf(i),Yf(i) = X and Y coordinates of the final end of the segment.

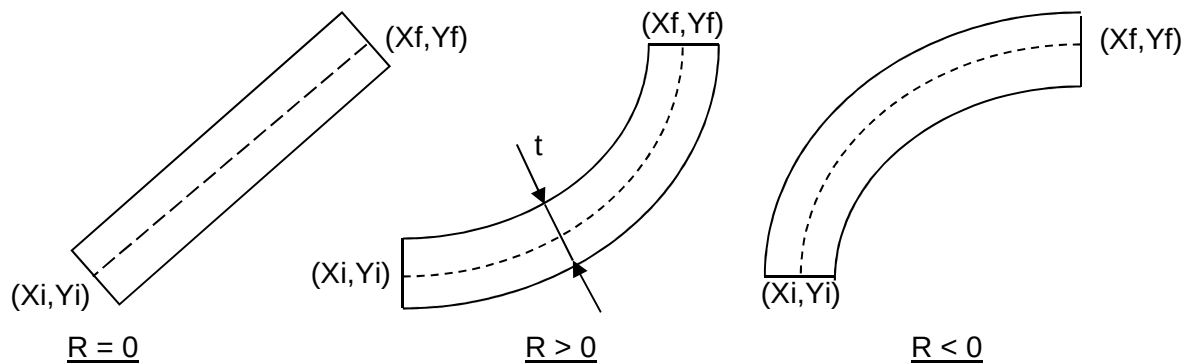
R(i) = Radius of the mid line of the segment, with the following conventions (see figure below):

R = 0: Straight segment.

R < 0: Circular arc with radius R in which the arc from the start point to the end point is in the clockwise direction.

R > 0: Circular arc with radius R in which the arc from the start point to the end point is in the anticlockwise direction.

t(i) = Thickness of the segment.



For circular arc segments, initial and final points must be located in such a way that the arc is no larger than 180° . If it is longer, it is necessary to divide the segment, establishing intermediate points.

The segments do not need to be specified in any particular order.

GROUP 2: Section defined by points of the contour line:

This group contains the geometrical definition of the cross section of the beam when it is defined by points of the contour line.

As pointed before, if the group 2 is present, group 1 have to be omitted, but one of them must be necessarily included in the input data file.

The structure of this group is the following:

Line 1 -----> 2 "Optional commentary"

[Np] lines ----->	Np	X(1)	Y(1)	R(1)
		X(2)	Y(2)	R(2)
				
				
		X(Np)	Y(Np)	R(Np)

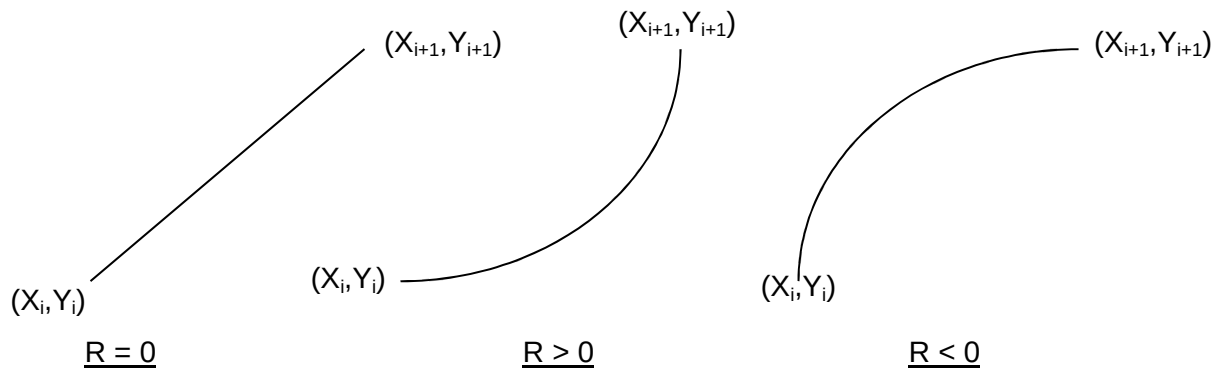
Where: Np = Number of points to define the contour of the section.

X(i),Y(i) = X and Y coordinates of the point.

R(i) = Radius of the arc joining the point "i" and the point "i+1", with the following conventions (see figure below):

R = 0: Straight line.

- $R < 0$: Circular arc with radius R in which the arc from the point "i" to the point "i+1" is in the clockwise direction.
- $R > 0$: Circular arc with radius R in which the arc from the point "i" to the point "i+1" is in the anticlockwise direction.



For circular arc lines, the end points must be defined so that the arc is no larger than 180° . Otherwise the arc must be split into subsections by selecting intermediate points.

In this case the contour lines must be in the correct order, since the program "builds up" the section by joining each point to the start point of the next segment according to the order in which they were provided to the program.

GROUP 3: Material properties:

This group contains the mechanical properties of the material of the beam. Its arrangement is the following:

Line 1 -----> 3 "Optional commentary"

Line 2 -----> E_t E_c F_{ty} F_{cy} F_{tu} e_u

Where: E_t = Tensile elastic modulus.

E^c = Compression elastic modulus.

F_{ty} = Tensile yield stress.

F_{cy} = Compression yield stress.

F_{tu} = Ultimate tensile stress.

e_u = Maximum plastic elongation, expressed in percentage (%).

This group can be omitted, only if group 4 is also omitted. In such a case, the flexion analysis will not be carried out and only the geometrical properties of the cross section will be calculated.

GROUP 4: Load cases:

This group contains the definition of the load cases to be analysed. The data structure in this group is as follows:

Line 1 -----> 4 "Optional commentary"

[Nc] lines ----->	Nc	Fz(1)	Mx(1)	My(1)	[Xo(1)]	[Yo(1)]
		Fz(2)	Mx(2)	My(2)	[Xo(2)]	[Yo(2)]
						
						
		Fz(Nc)	Mx(Nc)	My(Nc)	[Xo(Nc)]	[Yo(Nc)]

Where: Nc = Number of load cases.

Fz(i) = Axial force in the load case "i".

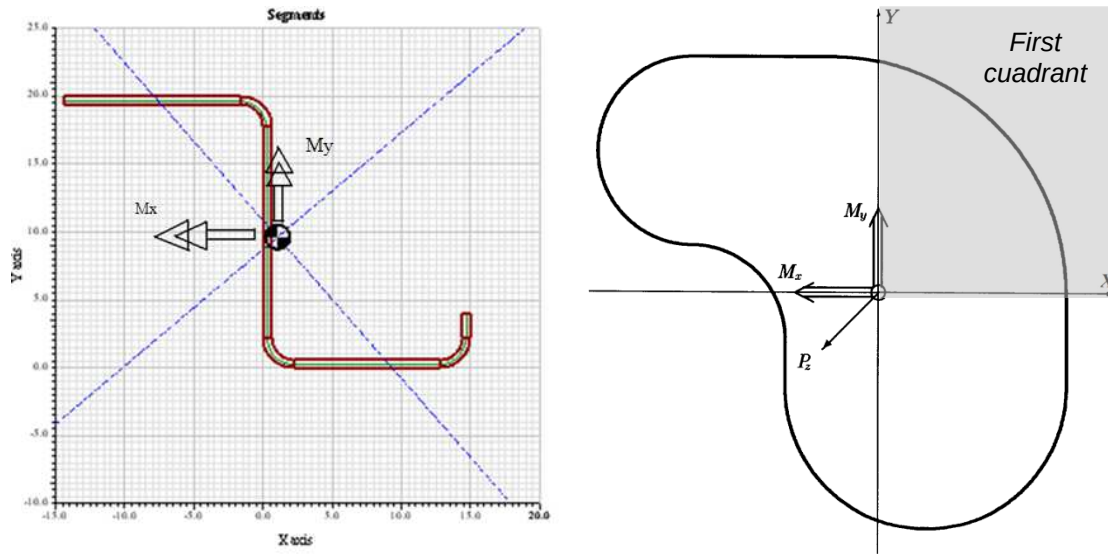
Mx(i), My(i) = Bending moment in X and Y directions for the load case "i".

Xo(i), Yo(i) = Coordinates of the application point of the load in the case "i".

If Xo(i) and Yo(i) are omitted, it will be assumed that the load in case "i" is applied on the centre of gravity of the section.

This group is optional. If it is omitted, the flexion analysis will not be carried out. In that case, only the geometrical properties of the cross section will be calculated.

Bending moment in X and Y directions have positive sign when causing compression in the first quadrant (area with x and y coordinates positive), see next figure. Axial force in Z direction is positive if produces tension in the beam.



GROUP 5: Maximum allowable strains:

This group contains the allowable strains under tension and compression for which the reserve factors will be calculated. The organization of this group is the following:

Line 1 -----> 5 "Optional commentary"

Line 2 -----> EpsT EpsC

Where: EpsT = Maximum allowable tensile microstrains.

EpsC = Maximum allowable compression microstrains.

This group is optional and it can be omitted. In such a case, the maximum strain capability of the material will be assumed, calculated according to the following expressions:

$$EpsT = 10^6 \cdot \frac{F_{tu}}{E_t} + 10^4 \cdot e_u \quad ; \quad EpsC = 10^6 \cdot \frac{F_{tu}}{E_c} + 10^4 \cdot e_u$$

For material with $F_{cy} > F_{tu}$, the maximum strain capability for compression "EpsC" is calculated as follows:

$$EpsC = 10^6 \cdot \frac{F_{cy}}{E_c} + 10^4 \cdot e_u$$

GROUP 6: Units:

This group contains the force and length units to be used by the program. It is arranged as follows:

Line 1 -----> 6 "Optional commentary"

Line 2 -----> [Force]

Line 3 -----> [Length]

Where: [Force] = Force units. It can be one of the following strings:
N, daN, kg, lb, klb.

[Length] = length units. It can be one of the following strings:
mm, cm, in.

This group is also optional and it can be omitted. In such a case, the following units will be assumed:

[Force] = N

[Length] = mm

4 OUTPUT RESULTS

4.1 WINDOWS PC/GUI

Pressing the button “*Flexion Analysis*” at the input data application form (see sub-chapter 3.1), the program performs the analysis and provides the output results by means of the following forms (data corresponding to the Example 1, see chapter 6):

Flexa 3.4 - ejemplo1.dat

Results for applied loads

Results for failure loads

	Loads			Neutral axis		Points of Maximum strain								R.F.
	Fz	Mx	My	Ang (°)	Dist. CG	X	Y	MicroStr.	Stress	X	Y	MicroStr.	Stress	
1	-10000.0	10000.0	0.0	137.5	-6.477	-0.110	19.419	-56830	-397.35	0.710	0.581	1893	137.04	1.13
2	0.0	10000.0	0.0	138.2	0.029	-0.110	19.419	-1611	-118.79	0.710	0.581	1738	125.79	>2
3	10000.0	10000.0	0.0	137.2	6.115	-0.110	19.419	-688	-50.74	0.710	0.581	32140	396.96	1.14
4	-10000.0	0.0	10000.0	126.6	-5.513	15.000	4.000	-80459	-414.46	-14.400	19.400	6376	328.93	1.08
5	0.0	0.0	10000.0	125.1	0.026	15.000	4.000	-2342	-170.73	-14.400	19.400	2024	146.53	>2
6	10000.0	0.0	10000.0	129.9	5.866	15.000	4.000	-3332	-223.27	0.464	0.849	32648	397.48	1.13
7	-10000.0	10000.0	10000.0	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	0.98
8	0.0	10000.0	10000.0	131.1	-0.008	15.000	4.000	-3697	-235.94	0.581	0.710	3361	241.70	>2
9	10000.0	10000.0	10000.0	133.1	4.858	15.000	4.000	-16962	-338.52	0.581	0.710	128564	441.82	1.01



Return to data panel

Exit from the program

Flexa 3.4 - ejemplo1.dat

Results for applied loads

Results for failure loads

	Loads			Neutral axis		Points of Maximum strain								R.F.
	Fz	Mx	My	Ang (°)	Dist. CG	X	Y	MicroStr.	Stress	X	Y	MicroStr.	Stress	
1	-11337.1	11337.1	0.0	137.8	-6.415	-0.110	19.419	-156075	-448.16	0.710	0.581	6232	327.34	1.13
2	0.0	53516.1	0.0	139.0	-0.034	-0.110	19.419	-148189	-445.46	0.710	0.581	156190	448.16	>2
3	11418.2	11418.2	0.0	136.0	5.598	-0.110	19.419	-8041	-298.15	0.710	0.581	156190	448.16	1.14
4	-10857.4	0.0	10857.4	126.8	-5.438	15.000	4.000	-156075	-448.16	-14.400	19.400	12584	363.36	1.08
5	0.0	0.0	45876.5	121.7	-0.269	15.000	4.000	-156075	-448.16	-14.400	19.400	131641	442.59	>2
6	11370.1	0.0	11370.1	131.3	5.401	15.000	4.000	-18225	-342.10	0.581	0.710	156190	448.16	1.13
7	-9840.3	9840.3	9840.3	128.8	-4.940	15.000	4.000	-156075	-448.16	0.464	0.849	13026	364.75	0.98
8	0.0	26868.4	26868.4	131.3	-0.124	15.000	4.000	-156075	-448.16	0.581	0.710	138879	444.33	>2
9	10152.5	10152.5	10152.5	133.1	4.860	15.000	4.000	-20565	-348.07	0.581	0.710	156190	448.16	1.01

Return to data panel

Exit from the program



This button, located at the lower left frame, is to write all the output results in a text file.

Values shown in output forms have the same meanings as in the output text files. Detailed explanations can be found in following sub-chapter.

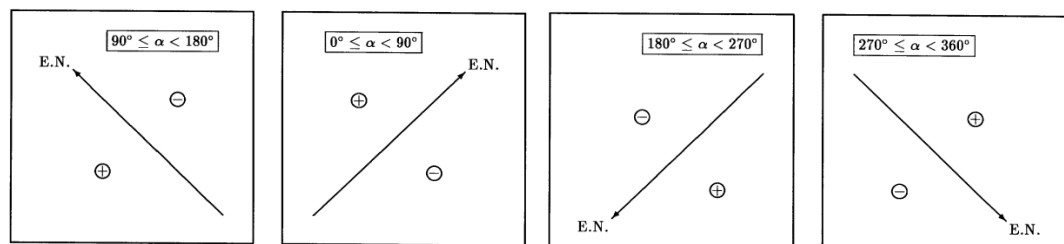
4.2 OUTPUT FILE

This file contains the following information:

- Input data:
 - Method to define the geometry.
 - Coordinates and parameters defining the geometry of the section.
 - Material data.
- Section properties:
 - Area of the section.
 - Centre of gravity position.
 - Moments of inertia in the centre of gravity, relative to the user defined reference axis system, and also in principal axes.
 - Angle between reference axes and principal axes.
- Analysis results:

For each load case, the following results are provided:

- Results for applied loads:
 - = Applied loads (input data).
 - = Neutral axis: Orientation and distance to the centre of gravity of the section (see figure below).
 - = Maximum deformation points: Coordinates (X,Y), strain and stress for points with maximum tension and compression strains (positive sign of stress/strain if the material is in tension).



Sign conventions for the distance of the neutral line to the centre of gravity.

- Results for failure loads:
 - = Failure loads: Factorized applied loads.
 - = Neutral axis: Orientation and distance to the centre of gravity of the section.

- = Maximum deformation points: Coordinates (X,Y), strain and stress for points with maximum tension and compression strains.
- Reserve factor value.

If the reserve factor is less than unity, neither the neutral axis nor the maximum deformation points are provided because the beam fails under applied loads.

Note that the neutral axis for applied loads is different from the corresponding to failure loads due to plasticity effects.

4.3 WARNING AND ERROR MESSAGES

In console versions, when the program detects an error in the input data or writing the results, it writes the corresponding message in the console and aborts the execution of the program.

The error messages that might appear in the console are listed below:

English output error messages:

- ERROR #1: The input file does not exist.
- ERROR #2: The input file cannot be opened.
- ERROR #3: Reading error in the input file.
- ERROR #4: End of file while reading input data.
- ERROR #5: Erroneous number of group data.
- ERROR #6: Erroneous number of segments to define the section.
- ERROR #7: Too many segments to define the section, exceeding the program capability.
- ERROR #8: Erroneous number of points to define the section.
- ERROR #9: Too many points to define the section, exceeding the program capability.
- ERROR #10: The material properties are inconsistent.
- ERROR #11: Erroneous number of load cases.
- ERROR #12: Too many load cases, exceeding the program capability.
- ERROR #13: The allowable strain in tension cannot be negative.
- ERROR #14: The section properties cannot be calculated.
- ERROR #15: The equivalent section cannot be calculated.

ERROR #16: The output file cannot be opened.

ERROR #17: Writing error in the output file.

ERROR #18: Defined points are coincident in plane.

ERROR #19: Defined corner radius too short.

Spanish output error messages:

ERROR #1: No existe el fichero de entrada.

ERROR #2: No se puede abrir el fichero de entrada.

ERROR #3: Error de lectura en el fichero de entrada.

ERROR #4: Fin de fichero mientras se leían los datos de entrada.

ERROR #5: Número de grupo de datos erróneo.

ERROR #6: Número erróneo de tramos para definir la sección.

ERROR #7: Demasiados tramos para definir la sección, excediendo la capacidad del programa.

ERROR #8: Número erróneo de puntos para definir la sección.

ERROR #9: Demasiados puntos para definir la sección, excediendo la capacidad del programa.

ERROR #10: Las propiedades del material son inconsistentes.

ERROR #11: Número erróneo de casos de carga.

ERROR #12: Demasiados casos de carga, excediendo la capacidad del programa.

ERROR #13: La deformación admisible a tracción no puede ser negativa.

ERROR #14: No se pueden calcular las propiedades de la sección.

ERROR #15: No se puede calcular la sección equivalente.

ERROR #16: No se puede abrir el fichero de salida.

ERROR #17: Error de escritura en el fichero de salida.

ERROR #18: Puntos definidos coincidentes en el plano.

ERROR #19: Radio de acuerdo definido demasiado pequeño.

5 LIMITATIONS

The FLEXA program analyses a beam element subjected to bending loads, or to a combination of bending and axial loads, and whose material works under elastic or plastic conditions.

This program can be used to analyse beams that fulfil with the following conditions:

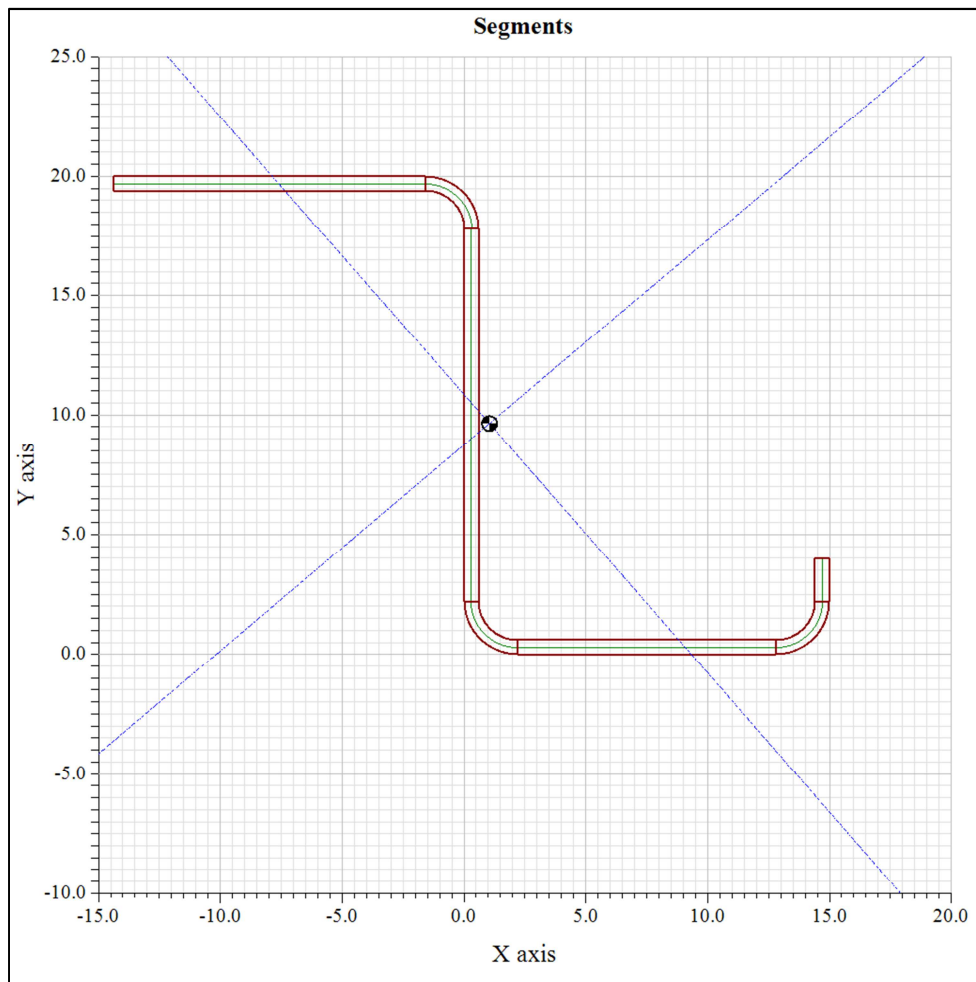
- 1) The cross-section of the beam can take any form (open or closed, thin-walled or otherwise), provided that its contour consists exclusively of straight lines and/or circular arcs.
- 2) The beam must be made of homogenous material, although this may have different characteristics for tensile and compressive loads.
- 3) Shear strains must be negligible. This is equivalent to say that plane cross-sections remain plane when deformed. This hypothesis is valid when the length of the beam is very much greater than the height of its cross-section (for example, for square - or "I"- section beams, a value of $L/h > 10$ is normally used).
- 4) The section should be stable, i.e., no part of the section should be subject to local buckling. Therefore, the results given by the program are only valid if the stresses produced in each segment of the section are less than the stresses required to cause local buckling.

6 EXAMPLES

In the following examples, the geometrical properties calculation as well as the flexion analysis is performed for four different sections. In the first and the second examples, the sections are defined by segments meanwhile in the other two examples, are defined by points of the contour line.

6.1 EXAMPLE 1

This cross-section has been defined by segments and it has the following geometry:



The material properties are the following:

$$\begin{aligned}
 E_t &= 72394.6 \text{ N/mm}^2 \\
 E_c &= 73773.6 \text{ N/mm}^2 \\
 F_{ty} &= 330.95 \text{ N/mm}^2 \\
 F_{cy} &= 275.79 \text{ N/mm}^2 \\
 F_{tu} &= 448.16 \text{ N/mm}^2 \\
 e_u &= 15.0 \%
 \end{aligned}$$

Nine load cases have been defined as shown on the table below. The axial load F_z is supposed applied on the centre of gravity of the section.

Case	F_z (N)	M_x (N·mm)	M_y (N·mm)
1	-10000.0	10000.0	0.0
2	0.0	10000.0	0.0
3	10000.0	10000.0	0.0
4	-10000.0	0.0	10000.0
5	0.0	0.0	10000.0
6	10000.0	0.0	10000.0
7	-10000.0	10000.0	10000.0
8	0.0	10000.0	10000.0
9	10000.0	10000.0	10000.0

The maximum allowable strains are supposed to be the corresponding to failure. They can be then calculated and they are the following ones:

Maximum allowable tensile strain = 156190.5 $\mu\epsilon$

Maximum allowable compression strain = 156074.8 $\mu\epsilon$

The associated input data file as well as the output results file are the following ones:

Input file:

```

1 Section defined by segments: N / Xi Yi Xf Yf R t
7 -14.4 19.7 -1.6 19.7 0.0 0.6
  -1.6 19.7 0.3 17.8 -1.9 0.6
  0.3 17.8 0.3 2.2 0.0 0.6
  0.3 2.2 2.2 0.3 1.9 0.6
  2.2 0.3 12.8 0.3 0.0 0.6
  12.8 0.3 14.7 2.2 1.9 0.6
  14.7 2.2 14.7 4.0 0.0 0.6
3 Material properties: Et, Ec, Fty, Fcy, Ftu, e(%)
72394.6 73773.6 330.95 275.79 448.16 15.0
4 Load cases: Fz Mx My [Xo Yo]
9 -10000.0 10000.0 0.0
  0.0 10000.0 0.0
  10000.0 10000.0 0.0
  -10000.0 0.0 10000.0
  0.0 0.0 10000.0
  10000.0 0.0 10000.0
  -10000.0 10000.0 10000.0
  0.0 10000.0 10000.0
  10000.0 10000.0 10000.0
5 Allowable microstrains: Eps-t Eps-c
156190.5 156074.8
6 Units: [Force] / [Length]
N
mm
  
```


Output file (Windows/PC console):

```

FLEXA 3.4                      (c) Airbus DS - 2016                      15/Feb/2016
-----

*****
*                               *
*   <<< Section definition >>>   *
*                               *
*****

Section defined by segments.

  Seg      Xi      Yi      Xf      Yf      R      t
-----
  1      -14.400    19.700    -1.600    19.700    0.0000    0.6000
  2      -1.600    19.700     0.300    17.800   -1.9000    0.6000
  3       0.300    17.800     0.300     2.200    0.0000    0.6000
  4       0.300     2.200     2.200     0.300    1.9000    0.6000
  5       2.200     0.300     12.800     0.300    0.0000    0.6000
  6      12.800     0.300    14.700     2.200    1.9000    0.6000
  7      14.700     2.200    14.700     4.000    0.0000    0.6000

Units: mm

Area of the section:
-----
A = 29.852 mm²

Center of gravity position:
-----

Xcg = 1.043 mm
Ycg = 9.638 mm

Moments of inertia in the center of gravity:
-----

Ixx = 1998.1 mm4
Iyy = 1571.7 mm4
Pxy = -1405.9 mm4

Moments of inertia in principal axes:
-----

I1 = 3206.8 mm4
I2 = 362.9 mm4

Ang = 40.69 °

*****
*                               *
*   <<< Material data >>>   *
*                               *
*****

                Tension      Compression
                -----
Modulus of elasticity = 72394.6      73773.6 N/mm²
Yield stress          = 330.95      275.79 N/mm²
Ultimate stress       = 448.16      N/mm²
Maximum elongation(%) = 15.00 %
No. of Ramberg-Osgood = 14.24      8.89
Allowable microstrain = 156190      156074
  
```


Output file, cont'd (Windows/PC console):

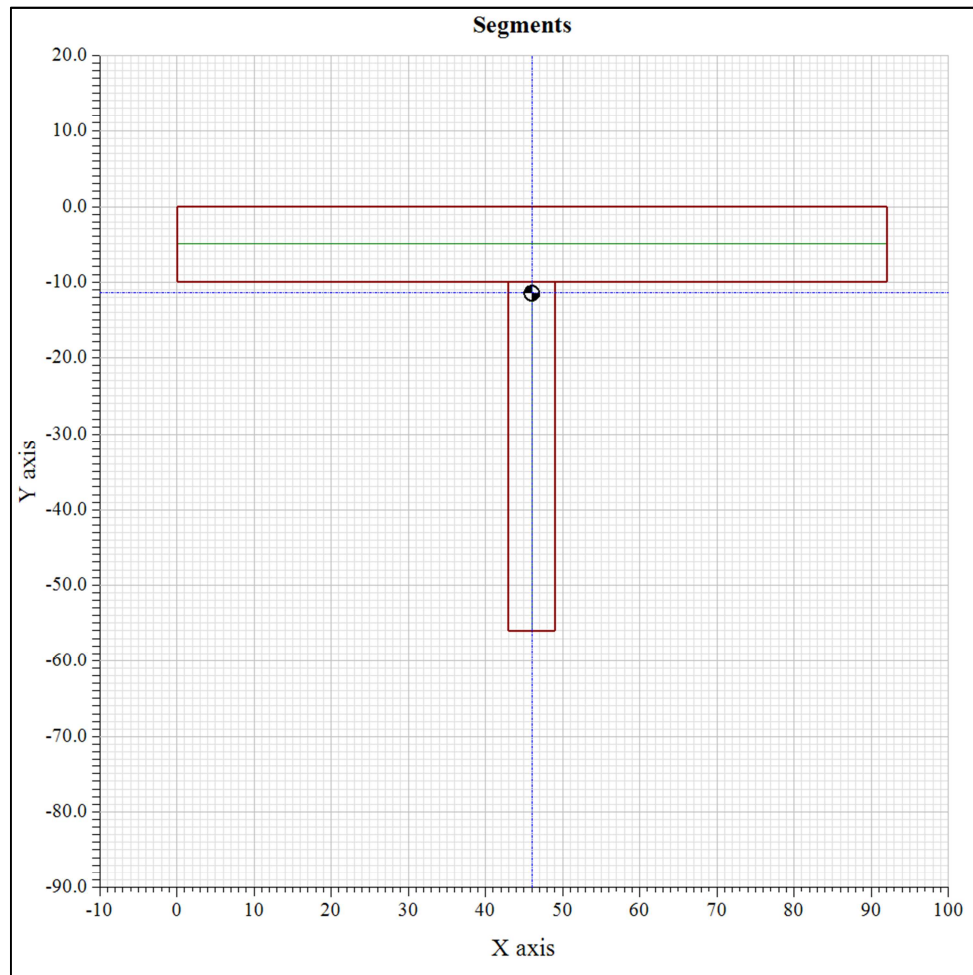
* <<< Análisis results >>> *													

Load case	Results for applied loads						Results for failure loads						R.F.
	Loads Pz Mx My	Neutral Axis Angle OX Dist. CG	Maximum deformation				Loads Pz Mx My	Neutral Axis Angle OX Dist. CG	Maximum deformation				
			Location	X,Y (User axes)	Strain	Stress			Location	X,Y (User axes)	Strain	Stress	
1	-10000.0 10000.0 0.0	137.5° -6.5	-0.110 0.710	19.419 0.581	-56830 1893	-397.35 137.04	-11337.1 11337.1 0.0	137.8° -6.4	-0.110 0.710	19.419 0.581	-156075 6232	-448.16 327.34	1.13
2	0.0 10000.0 0.0	138.2° 0.0	-0.110 0.710	19.419 0.581	-1611 1738	-118.79 125.79	0.0 53516.1 0.0	139.0° -0.0	-0.110 0.710	19.419 0.581	-148189 156191	-445.46 448.16	5.35
3	10000.0 10000.0 0.0	137.2° 6.1	-0.110 0.710	19.419 0.581	-688 32140	-50.74 396.96	11418.2 11418.2 0.0	136.0° 5.6	-0.110 0.710	19.419 0.581	-8041 156191	-298.14 448.16	1.14
4	-10000.0 0.0 10000.0	126.6° -5.5	15.000 -14.400	4.000 19.400	-80459 6376	-414.46 328.93	-10857.4 0.0 10857.4	126.8° -5.4	15.000 -14.400	4.000 19.400	-156075 12584	-448.16 363.36	1.08
5	0.0 0.0 10000.0	125.1° 0.0	15.000 -14.400	4.000 19.400	-2342 2024	-170.73 146.53	0.0 0.0 45876.5	121.7° -0.3	15.000 -14.400	4.000 19.400	-156075 131641	-448.16 442.59	4.58
6	10000.0 0.0 10000.0	129.9° 5.9	15.000 0.464	4.000 0.849	-3332 32648	-223.27 397.48	11370.1 0.0 11370.1	131.3° 5.4	15.000 0.581	4.000 0.710	-18225 156191	-342.10 448.16	1.13
7	-10000.0 10000.0 10000.0	----- -----	----- -----	----- -----	----- -----	----- -----	-9840.3 9840.3 9840.3	128.8° -4.9	15.000 0.464	4.000 0.849	-156075 13026	-448.16 364.75	0.98
8	0.0 10000.0 10000.0	131.1° -0.0	15.000 0.581	4.000 0.710	-3697 3361	-235.94 241.70	0.0 26868.4 26868.4	131.3° -0.1	15.000 0.581	4.000 0.710	-156075 138879	-448.16 444.33	2.68
9	10000.0 10000.0 10000.0	133.1° 4.9	15.000 0.581	4.000 0.710	-16962 128564	-338.52 441.82	10152.5 10152.5 10152.5	133.1° 4.9	15.000 0.581	4.000 0.710	-20565 156191	-348.07 448.16	1.01

Units: N, mm													
All loads are referred to the center of gravity													
Load case with minimum reserve factor = 7													
RFmin = 0.98													

6.2 EXAMPLE 2

This cross-section has been defined by segments and it has the following geometry:



The material properties are the following:

$$\begin{aligned}
 E_t &= 71705.0 \text{ N/mm}^2 \\
 E_c &= 73773.6 \text{ N/mm}^2 \\
 F_{ty} &= 434.37 \text{ N/mm}^2 \\
 F_{cy} &= 434.37 \text{ N/mm}^2 \\
 F_{tu} &= 510.21 \text{ N/mm}^2 \\
 e_u &= 8.0 \%
 \end{aligned}$$

Nine load cases have been defined as shown on the table below. The axial load F_z is supposed applied on the centre of gravity of the section.

Case	Fz (N)	Mx (N·mm)	My (N·mm)
1	-400000.0	400000.0	0.0
2	0.0	400000.0	0.0
3	400000.0	400000.0	0.0
4	-400000.0	0.0	400000.0
5	0.0	0.0	400000.0
6	400000.0	0.0	400000.0
7	-400000.0	400000.0	400000.0
8	0.0	400000.0	400000.0
9	400000.0	400000.0	400000.0

The maximum allowable strains are supposed to be the corresponding to failure. They can be then calculated and they are the following ones:

Maximum allowable tensile strain = 87115.41 $\mu\epsilon$

Maximum allowable compression strain = 86915.89 $\mu\epsilon$

The associated input data file as well as the output results file are the following ones:

Input file:

```

1 Section defined by segments: N / Xi Yi Xf Yf R t
2 0.0 -5.0 92.0 -5.0 0.00 10.0
   46.0 -10.0 46.0 -56.0 0.00 6.0
3 Material properties: Et, Ec, Fty, Fcy, Ftu, e(%)
   71705.0 73773.6 434.37 434.37 510.21 8.0
4 Load cases: Fz Mx My [Xo Yo]
9 -400000.0 400000.0 0.0
   0.0 400000.0 0.0
   400000.0 400000.0 0.0
  -400000.0 0.0 400000.0
   0.0 0.0 400000.0
   400000.0 0.0 400000.0
  -400000.0 400000.0 400000.0
   0.0 400000.0 400000.0
   400000.0 400000.0 400000.0
5 Allowable microstrains: Eps-t Eps-c
   87115.41 86915.89
6 Units: [Force] / [Length]
   N
   mm
  
```

Output file (Windows/PC console):

FLEXA 3.4

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<<< Section definition >>>

Section defined by segments.

Seg	Xi	Yi	Xf	Yf	R	t
1	0.000	-5.000	92.000	-5.000	0.00	10.000
2	46.000	-10.000	46.000	-56.000	0.00	6.000

Units: mm

Area of the section:

A = 1196.0 mm²

Center of gravity position:

Xcg = 46.000 mm

Ycg = -11.462 mm

Moments of inertia in the center of gravity:

Ixx = 222783.9 mm4

Iyy = 649734.7 mm4

Pxy = 0.0 mm4

Moments of inertia in principal axes:

I1 = 222783.9 mm4

I2 = 649734.7 mm4

Ang = 0.00 °

<<< Material data >>>

	Tension	Compression	
Modulus of elasticity =	71705.0	73773.6	N/mm²
Yield stress =	434.37	434.37	N/mm²
Ultimate stress =	510.21		N/mm²
Maximum elongation(%) =	8.00 %		
No. of Ramberg-Osgood =	22.92	22.92	
Allowable microstrain =	87115	86915	

<<< Analisis results >>>

Results for applied loads						Results for failure loads							
Load case	Loads	Neutral Axis	Maximum deformation				Loads	Neutral Axis	Maximum deformation				R.F.
	Pz	Angle OX	Location X,Y (User axes)	Strain	Stress	Pz	Angle OX	Location X,Y (User axes)	Strain	Stress			
	Mx	Dist. CG				Mx	Dist. CG						
	My					My							
1	-400000.0	180.0°	92.000 0.000	-4825	-354.58	-591659.1	180.0°	92.000 0.000	-86916	-510.21	1.47		
	400000.0	-184.1	43.000 -56.000	-3443	-254.02	591659.1	-43.5	43.000 -56.000	1562	112.03			
	0.0					0.0							

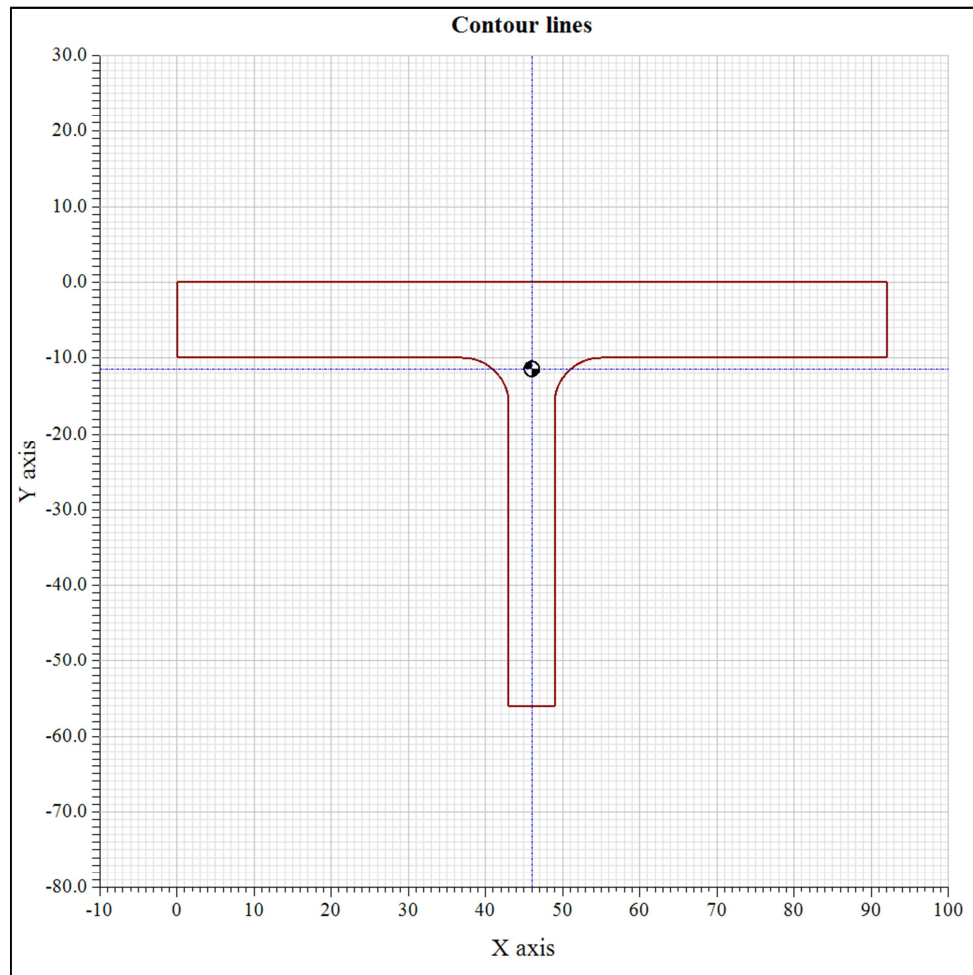
Output file, cont'd (Windows/PC console):

2	0.0	180.0°	92.000	0.000	-282	-20.79	0.0	180.0°	92.000	0.000	-11745	-454.28	11.4
	400000.0	0.1	43.000	-56.000	1112	79.75	4581350.5	4.8	43.000	-56.000	87115	510.21	
	0.0						0.0						
3	400000.0	180.0°	92.000	0.000	4350	311.87	532944.7	180.0°	92.000	0.000	3432	246.12	1.33
	400000.0	156.6	43.000	-56.000	6029	404.43	532944.7	13.8	43.000	-56.000	87115	510.21	
	0.0						0.0						
4	-400000.0	90.5°	92.000	0.000	-4939	-362.08	-580244.8	110.4°	92.000	0.000	-86916	-510.21	1.45
	0.0	-526.3	0.000	-10.000	-4144	-305.70	0.0	-44.9	0.000	-10.000	-2130	-157.17	
	400000.0						580244.8						
5	0.0	89.6°	92.000	-10.000	-387	-28.56	0.0	90.0°	92.000	-10.000	-86916	-510.21	26.5
	0.0	0.3	0.000	0.000	392	28.13	0.0	0.0	0.000	0.000	87005	510.18	
	400000.0						10626299.0						
6	400000.0	89.5°	92.000	-10.000	4264	305.71	580226.9	69.6°	92.000	-10.000	2269	162.67	1.45
	0.0	526.8	0.000	0.000	5081	362.10	0.0	45.0	0.000	0.000	87115	510.21	
	400000.0						580226.9						
7	-400000.0	160.2°	92.000	0.000	-5244	-380.01	-565467.9	135.3°	92.000	0.000	-86916	-510.21	1.41
	400000.0	-172.0	43.000	-56.000	-3412	-251.75	565467.9	-31.9	43.000	-56.000	2205	158.13	
	400000.0						565467.9						
8	0.0	161.3°	92.000	0.000	-667	-49.21	0.0	174.9°	92.000	0.000	-19413	-471.36	11.1
	400000.0	0.2	43.000	-56.000	1137	81.55	4452154.5	4.6	43.000	-56.000	87115	510.21	
	400000.0						4452154.5						
9	400000.0	163.8°	92.000	0.000	3951	283.28	532953.3	177.6°	92.000	0.000	1040	74.56	1.33
	400000.0	150.5	43.000	-56.000	6055	405.14	532953.3	14.1	43.000	-56.000	87115	510.21	
	400000.0						532953.3						

Units: N, mm													
All loads are referred to the center of gravity													
Load case with minimum reserve factor = 3													
RFmin = 1.33													

6.3 EXAMPLE 3

This cross-section has been defined by segments and it has the following geometry:



The material properties are the following:

$$\begin{aligned}
 E_t &= 71705.0 \text{ N/mm}^2 \\
 E_c &= 73774.0 \text{ N/mm}^2 \\
 F_{ty} &= 434.37 \text{ N/mm}^2 \\
 F_{cy} &= 434.37 \text{ N/mm}^2 \\
 F_{tu} &= 510.21 \text{ N/mm}^2 \\
 e_u &= 8.0 \%
 \end{aligned}$$

Nine load cases have been defined as shown on the table below. The axial load F_z is supposed applied on the centre of gravity of the section.

Case	Fz (N)	Mx (N·mm)	My (N·mm)
1	-400000.0	400000.0	0.0
2	0.0	400000.0	0.0
3	400000.0	400000.0	0.0
4	-400000.0	0.0	400000.0
5	0.0	0.0	400000.0
6	400000.0	0.0	400000.0
7	-400000.0	400000.0	400000.0
8	0.0	400000.0	400000.0
9	400000.0	400000.0	400000.0

The maximum allowable strains are supposed to be the corresponding to failure. They can be then calculated and they are the following ones:

Maximum allowable tensile strain = 87115.41 $\mu\epsilon$

Maximum allowable compression strain = 86915.85 $\mu\epsilon$

The associated input data file as well as the output results file are the following ones:

Input file:

```

2 Section defined by contour lines: N / X Y R
10  0.0  0.0  0.0
    92.0  0.0  0.0
    92.0 -10.0  0.0
    55.0 -10.0  6.0
    49.0 -16.0  0.0
    49.0 -56.0  0.0
    43.0 -56.0  0.0
    43.0 -16.0  6.0
    37.0 -10.0  0.0
    0.0 -10.0  0.0
3 Material properties: Et, Ec, Fty, Fcy, Ftu, e(%)
71705.0 73774.0 434.37 434.37 510.21 8.0
4 Load cases: Fz Mx My [Xo Yo]
9  -400000.0  400000.0  0.0
    0.0  400000.0  0.0
    400000.0  400000.0  0.0
   -400000.0  0.0  400000.0
    0.0  0.0  400000.0
    400000.0  0.0  400000.0
   -400000.0  400000.0  400000.0
    0.0  400000.0  400000.0
    400000.0  400000.0  400000.0
5 Allowable microstrains: Eps-t Eps-c
87115.41 86915.85
6 Units: [Force] / [Length]
N
mm
  
```

Output file (Windows/PC console):

FLEXA 3.4 (c) Airbus DS - 2016 15/Feb/2016

```

*****
*                                     *
*   <<< Section definition >>>   *
*                                     *
*****
  
```

Section defined by contour lines.

Point	X	Y	R
1	0.000	0.000	0.0000
2	92.000	0.000	0.0000
3	92.000	-10.000	0.0000
4	55.000	-10.000	6.0000
5	49.000	-16.000	0.0000
6	49.000	-56.000	0.0000
7	43.000	-56.000	0.0000
8	43.000	-16.000	6.0000
9	37.000	-10.000	0.0000
10	0.000	-10.000	0.0000

Units: mm

Area of the section:

 A = 1211.5 mm²

Center of gravity position:

Xcg = 46.000 mm

Ycg = -11.460 mm

Moments of inertia in the center of gravity:

 Ixx = 222803.8 mm⁴

 Iyy = 650045.1 mm⁴

 Pxy = 0.0 mm⁴

Moments of inertia in principal axes:

 I1 = 222803.8 mm⁴

 I2 = 650045.1 mm⁴

Ang = 0.00 °

```

*****
*                                     *
*   <<< Material data >>>   *
*                                     *
*****
  
```

	Tension	Compression	
Modulus of elasticity =	71705.0	73774.0	N/mm ²
Yield stress =	434.37	434.37	N/mm ²
Ultimate stress =	510.21		N/mm ²
Maximum elongation(%) =	8.00 %		
No. of Ramberg-Osgood =	22.92	22.92	
Allowable microstrain =	87115	86915	

Output file, cont'd (Windows/PC console):

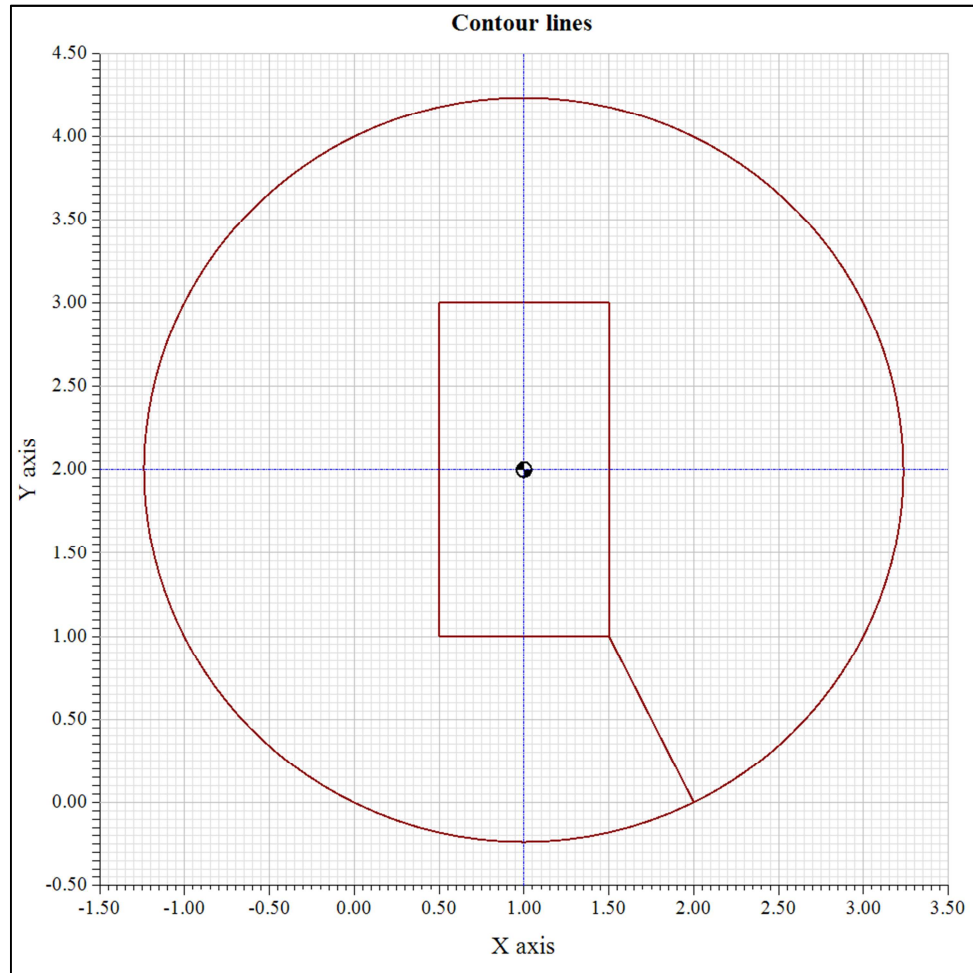
* <<< Analisis results >>> *												

Load case	Results for applied loads						Results for failure loads					
	Loads		Neutral Axis		Maximum deformation		Loads		Neutral Axis		Maximum deformation	
	Pz	Angle	OX	Location	X,Y	Strain, Stress	Pz	Angle	OX	Location	X,Y	Strain, Stress
	Mx	Dist. CG		(User axes)			Mx	Dist. CG		(User axes)		
	My						My					
1	-400000.0	180.0°	92.000	0.000	-4764	-350.39	-599300.6	180.0°	92.000	0.000	-86916	-510.21
	400000.0	-182.2	49.000	-56.000	-3387	-249.85	599300.6	-43.5	49.000	-56.000	1647	118.12
	0.0						0.0					
2	0.0	180.0°	92.000	0.000	-282	-20.79	0.0	180.0°	92.000	0.000	-11913	-454.84
	400000.0	0.1	49.000	-56.000	1112	79.75	4614524.0	4.7	49.000	-56.000	87115	510.21
	0.0						0.0					
3	400000.0	180.0°	92.000	0.000	4295	307.95	539589.1	180.0°	0.000	0.000	3382	242.48
	400000.0	159.6	49.000	-56.000	5919	401.20	539589.1	13.7	49.000	-56.000	87115	510.21
	0.0						0.0					
4	-400000.0	90.4°	92.000	0.000	-4876	-357.94	-587663.8	110.6°	92.000	0.000	-86916	-510.21
	0.0	-523.4	0.000	-10.000	-4087	-301.51	0.0	-44.7	0.000	-10.000	-2037	-150.30
	400000.0						587663.8					
5	0.0	89.6°	92.000	-10.000	-387	-28.55	0.0	90.0°	92.000	-10.000	-86916	-510.21
	0.0	0.3	0.000	0.000	392	28.11	0.0	0.0	0.000	0.000	87007	510.18
	400000.0						10655574.0					
6	400000.0	89.6°	92.000	-10.000	4206	301.54	587645.7	69.4°	92.000	-10.000	2175	155.96
	0.0	524.1	0.000	0.000	5016	357.94	0.0	44.8	0.000	0.000	87115	510.21
	400000.0						587645.7					
7	-400000.0	160.5°	92.000	0.000	-5174	-376.24	-572684.1	135.2°	92.000	0.000	-86916	-510.21
	400000.0	-170.8	43.000	-56.000	-3357	-247.68	572684.1	-31.9	43.000	-56.000	2272	162.91
	400000.0						572684.1					
8	0.0	161.4°	92.000	0.000	-667	-49.20	0.0	174.8°	92.000	0.000	-19747	-471.88
	400000.0	0.2	43.000	-56.000	1137	81.54	4484303.5	4.5	43.000	-56.000	87115	510.21
	400000.0						4484303.5					
9	400000.0	163.3°	92.000	0.000	3897	279.44	539576.2	177.6°	92.000	0.000	971	69.65
	400000.0	153.1	43.000	-56.000	5944	401.98	539576.2	14.0	43.000	-56.000	87115	510.21
	400000.0						539576.2					

Units: N, mm												
All loads are referred to the center of gravity												
Load case with minimum reserve factor = 9												
RFmin = 1.35												

6.4 EXAMPLE 4

This cross-section has been defined by segments and it has the following geometry:



The material properties are the following:

$$\begin{aligned}
 E_t &= 71705.0 \text{ N/mm}^2 \\
 E_c &= 73774.0 \text{ N/mm}^2 \\
 F_{ty} &= 434.37 \text{ N/mm}^2 \\
 F_{cy} &= 434.37 \text{ N/mm}^2 \\
 F_{tu} &= 510.21 \text{ N/mm}^2 \\
 e_u &= 8.0 \%
 \end{aligned}$$

Nine load cases have been defined as shown on the table below. The axial load F_z is supposed applied on the centre of gravity of the section.

Case	Fz (N)	Mx (N·mm)	My (N·mm)
1	-1000.0	1000.0	0.0
2	0.0	1000.0	0.0
3	1000.0	1000.0	0.0
4	-1000.0	0.0	1000.0
5	0.0	0.0	1000.0
6	1000.0	0.0	1000.0
7	-1000.0	1000.0	1000.0
8	0.0	1000.0	1000.0
9	1000.0	1000.0	1000.0

The maximum allowable strains are supposed to be the corresponding to failure. They can be then calculated and they are the following ones:

Maximum allowable tensile strain = 87115.41 $\mu\epsilon$

Maximum allowable compression strain = 86915.85 $\mu\epsilon$

The associated input data file as well as the output results file are the following ones:

Input file:

```

2 Section defined by contour lines: N / X Y R
10  0.0  0.0  2.2361
    2.0  0.0  0.0
    1.5  1.0  0.0
    0.5  1.0  0.0
    0.5  3.0  0.0
    1.5  3.0  0.0
    1.5  1.0  0.0
    2.0  0.0  2.2361
    2.0  4.0  2.2361
    0.0  4.0  2.2361
3 Material properties: Et, Ec, Fty, Fcy, Ftu, e(%)
71705.0 73774.0 434.37 434.37 510.21 8.0
4 Load cases: Fz Mx My [Xo Yo]
9  -1000.0 1000.0 0.0
    0.0 1000.0 0.0
    1000.0 1000.0 0.0
    -1000.0 0.0 1000.0
    0.0 0.0 1000.0
    1000.0 0.0 1000.0
    -1000.0 1000.0 1000.0
    0.0 1000.0 1000.0
    1000.0 1000.0 1000.0
5 Allowable microstrains: Eps-t Eps-c
87115.41 86915.85
6 Units: [Force] / [Length]
N
mm
  
```

Output file (Windows/PC console):

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 * <<< Section definition >>> *
 * *****

Section defined by contour lines.

Point	X	Y	R
1	0.0000	0.0000	2.2361
2	2.0000	0.0000	0.0000
3	1.5000	1.0000	0.0000
4	0.5000	1.0000	0.0000
5	0.5000	3.0000	0.0000
6	1.5000	3.0000	0.0000
7	1.5000	1.0000	0.0000
8	2.0000	0.0000	2.2361
9	2.0000	4.0000	2.2361
10	0.0000	4.0000	2.2361

Units: mm

Area of the section:

 A = 13.708 mm²

Center of gravity position:

Xcg = 1.0000 mm

Ycg = 2.0000 mm

Moments of inertia in the center of gravity:

 Ixx = 18.968 mm⁴
 Iyy = 19.467 mm⁴
 Pxy = 0.000 mm⁴

Moments of inertia in principal axes:

 I1 = 18.968 mm⁴
 I2 = 19.467 mm⁴

Ang = 0.00 °

 * <<< Material data >>> *
 * *****

	Tension	Compression	
Modulus of elasticity =	71705.0	73774.0	N/mm ²
Yield stress =	434.37	434.37	N/mm ²
Ultimate stress =	510.21		N/mm ²
Maximum elongation(%) =	8.00 %		
No. of Ramberg-Osgood =	22.92	22.92	
Allowable microstrain =	87115	86915	

Output file, cont'd (Windows/PC console):

* <<< Analisis results >>> *													

Load case	Results for applied loads						Results for failure loads						R.F.
	Loads Pz Mx My	Neutral Axis Angle OX Dist. CG	Maximum deformation				Loads Pz Mx My	Neutral Axis Angle OX Dist. CG	Maximum deformation				
			Location	X,Y (User axes)	Strain	Stress			Location	X,Y (User axes)	Strain	Stress	
1	-1000.0 1000.0 0.0	180.0° -1.4	1.0942 0.9058	4.2341 -0.2341	-2593 615	-191.28 44.07	-4195.6 4195.6 0.0	180.0° -1.2	1.0942 1.0942	4.2341 -0.2341	-86916 24807	-510.21 478.23	4.19
2	0.0 1000.0 0.0	180.0° 0.0	1.0942 0.9058	4.2341 -0.2341	-1613 1634	-118.99 117.16	0.0 6861.3 0.0	180.0° 0.0	1.0942 0.9058	4.2341 -0.2341	-86916 87015	-510.21 510.18	6.86
3	1000.0 1000.0 0.0	180.0° 1.4	1.0942 0.9058	4.2341 -0.2341	-623 2663	-45.99 190.96	4195.5 4195.5 0.0	180.0° 1.2	1.0942 0.9058	4.2341 -0.2341	-24747 87115	-478.37 510.21	4.19
4	-1000.0 0.0 1000.0	90.0° -1.4	3.2360 -1.2360	2.0000 2.0000	-2553 574	-188.33 41.18	-4197.4 0.0 4197.4	90.0° -1.2	3.2360 -1.2360	2.0000 2.0000	-86916 24740	-510.21 478.15	4.19
5	0.0 0.0 1000.0	90.0° 0.0	3.2360 -1.2360	2.0000 2.0000	-1573 1594	-116.02 114.28	0.0 0.0 7112.9	90.0° 0.0	3.2360 -1.2360	2.0000 2.0000	-86916 87019	-510.21 510.18	7.11
6	1000.0 0.0 1000.0	90.0° 1.4	3.2360 -1.2360	2.0000 2.0000	-583 2623	-43.02 188.05	4197.3 0.0 4197.3	90.0° 1.2	3.2360 -1.2360	2.0000 2.0000	-24678 87115	-478.29 510.21	4.19
7	-1000.0 1000.0 1000.0	135.7° -1.0	2.6107 -0.6107	3.5509 0.4491	-3232 1259	-238.45 90.30	-3514.6 3514.6 3514.6	135.1° -1.1	2.6107 -0.6107	3.5509 0.4491	-86916 31234	-510.21 484.52	3.51
8	0.0 1000.0 1000.0	135.7° 0.0	2.6107 -0.6107	3.5509 0.4491	-2253 2282	-166.18 163.65	0.0 4931.5 4931.5	136.9° 0.0	2.4730 -0.4730	3.6823 0.3177	-86916 87016	-510.21 510.18	4.93
9	1000.0 1000.0 1000.0	135.7° 1.0	2.6107 -0.6107	3.5509 0.4491	-1267 3312	-93.44 237.48	3514.6 3514.6 3514.6	135.1° 1.1	2.6107 -0.6107	3.5509 0.4491	-31179 87115	-484.63 510.21	3.51

Units: N, mm													
All loads are referred to the center of gravity													
Load case with minimum reserve factor = 9													
RFmin = 3.51													

6.5 EXAMPLE 5

The cross-section geometry of this example is the same as Example 4 (see sub-chapter 6.4), but the material in this case is a steel alloy with a yield compression stress greater than the tension ultimate stress. Two load cases have been defined as shown on the table below. The axial load F_z is supposed applied on the centre of gravity of the section.

Case	F_z (N)	M_x (N·mm)	M_y (N·mm)
1	-1000.0	10.0	0.0
2	0.0	1000.0	0.0

The associated input data file as well as the output results file are the following ones:

Input file:

```

2 Sección definida por líneas de contorno: N / X Y R
10 0.0000 0.0000 2.2361
   2.0000 0.0000 0.0000
   1.5000 1.0000 0.0000
   0.5000 1.0000 0.0000
   0.5000 3.0000 0.0000
   1.5000 3.0000 0.0000
   1.5000 1.0000 0.0000
   2.0000 0.0000 2.2361
   2.0000 4.0000 2.2361
   0.0000 4.0000 2.2361
3 Propiedades del material: Et, Ec, Fty, Fcy, Ftu, e(%)
  195121.0 202705.0 1379.0 1455.0 1434.0 10.0
4 Casos de carga: Fz Mx My [Xo Yo]
  2 -1000.0 10.0 0.0
    0.0 1000.0 0.0
5 Microdeformaciones admisibles: Eps-t Eps-c
  87115.41 86915.85
6 Unidades: [Fuerza] / [Longitud]
  N
  mm
  
```

Output file (Windows/PC console):

```

FLEXA 3.4                      (c) Airbus DS - 2016                      15/Feb/2016
-----
*****
*                               *
*   <<< Section definition >>>   *
*                               *
*****

Section defined by contour lines.

Point      X      Y      R
-----
  1      0.0000      0.0000      2.2361
  2      2.0000      0.0000      0.0000
  3      1.5000      1.0000      0.0000
  4      0.5000      1.0000      0.0000
  5      0.5000      3.0000      0.0000
  6      1.5000      3.0000      0.0000
  7      1.5000      1.0000      0.0000
  8      2.0000      0.0000      2.2361
  9      2.0000      4.0000      2.2361
 10      0.0000      4.0000      2.2361
  
```

Output file, cont'd (Windows/PC console):

```

Units: mm

Area of the section:
-----
A = 13.708 mm²

Center of gravity position:
-----

Xcg = 1.0000 mm
Ycg = 2.0000 mm

Moments of inertia in the center of gravity:
-----

Ixx = 18.968 mm⁴
Iyy = 19.467 mm⁴
Pxy = 0.000 mm⁴

Moments of inertia in principal axes:
-----

I1 = 18.968 mm⁴
I2 = 19.467 mm⁴

Ang = 0.00 °

*****
*                                     *
*   <<< Material data >>>         *
*                                     *
*****

                Tension      Compression
            -----
Modulus of elasticity = 195121.0      202705.0 N/mm²
Yield stress          = 1379.0        1455.0 N/mm²
Ultimate stress       = 1434.0        N/mm²
Maximum elongation(%) = 10.00 %
No. of Ramberg-Osgood = 100.03      *****

"For these cases (Fcy>Ftu) the material in compression is assumed perfectly-plastic after yield"
"The strength cut-off is set at the point with the same permanent deformation after failure as in tension."

Allowable microstrain =      87115      107177

*****
*                                     *
*   <<< Analysis results >>>       *
*                                     *
*****

                Results for applied loads
            -----
Load case   Loads      Neutral      Maximum deformation
            Pz      Axis      Location X,Y      Strain, Stress
            Mx      Angle OX      (User axes)
            My      Dist. CG
            -----
1           -1000.0  180.1°  0.9058  4.2341  -366  -74.1
            10.0    -151.5  1.0942 -0.2341  -355  -72.0
            0.0
2           0.0     180.0°  0.9058  4.2341  -589  -119.3
            1000.0  0.0    1.0942 -0.2341  599   116.8
            0.0

            -----

                Results for failure loads
            -----
Load case   Loads      Neutral      Maximum deformation
            Pz      Axis      Location X,Y      Strain, Stress
            Mx      Angle OX      (User axes)
            My      Dist. CG
            -----
1           -19825.1  180.0°  1.0942  4.2341  -107178 -1455.0
            198.3    -2.3   0.9058 -0.2341  -2565  -519.9
            0.0
2           0.0     180.0°  1.0942  4.2341  -85061 -1455.0
            19899.0  0.0   0.9058 -0.2341  87115  1430.8
            0.0

            -----
R.F.
19.8
19.8

Units: N, mm

All loads are referred to the center of gravity

Load case with minimum reserve factor = 1
                                         RFmin = 19.83
  
```